

Module 13

Hashing, HashMap and HashSet in Java

Last time:

We implemented a very simple HashTable using linear probing. Its hash function is also simple since the keys are positive integers and we assumed that it would be random.

```
key % tableLength;
```

But, having random keys is not always the case. To handle non-random keys, we talked about a few general rules for a better hash function.

- **Don't Use Non-Data:** The keys should be squeezed as much as it could. For example, category code has to be changed to be from 0 to 15. Also, the checksum should be removed because it is a derived number from other information and does not add any more information.
- **Use All the Data:** Other than the non-data values, we need to use all the data values. Don't just use the first four digits, etc.
- **Use a Prime Number for the Modulo Base:** Which means the table array length should be a prime number. This can prevent clusters from happening. For example, if the table size is 50, then all the multiples of 50 in a dataset will be hashed into the same index.
- **Use Folding:** Another reasonable hash function involves breaking the keys into groups of digits and adding (mixing up) the groups.

Today, we will look into hashing in Java.

hashCode() method

The hashCode() method in Java is implemented in the Object class.

What does it mean?

Each class in Java inherits it!

Basically, the hashCode() method returns a numeric representation of an object.

```
Integer intObj = Integer.valueOf(17683);
String strObj = "17683";
System.out.println("hashCode for integer: " + intObj.hashCode());
System.out.println("hashCode for string: " + strObj.hashCode());
```

This will print

hashCode for integer: 17683

hashCode for string: _____

For example, hashCode method in the String class uses the following formula.

$$s.charAt(0) * 31^{(n-1)} + s.charAt(1) * 31^{(n-2)} + \dots + s.charAt(n-1)$$

For example,
"cats"

$a \dots z \rightarrow 97 \dots 122$

$A \dots Z \rightarrow 65 \dots 90$

Now, why 31?

Prime (good for wrap around)

→ Collision

116×31

$116 \times 32 - 116$

What does this tell us?

$110100 < 5 = 110100000000$
= 进制 左移5位

HashMap

The Map interface in Java is not an extension of the Collection interface. It starts off its own hierarchy.

The interface describes a mapping from keys to values, without duplicate keys, by definition.

As a result, maps provide a good way of searching for an object on the value of another.

Other than the fact that it is key and value mapping implementation, HashMap is a hashtable as we discussed so far.

Here are some code snippets from HashMap class (Java 7).

```
/**
 * The table, resized as necessary. Length MUST Always be a power
of two.
 */
.... Entry<K,V>[] table;

public HashMap() {
    this.loadFactor = DEFAULT_LOAD_FACTOR;
    threshold = (int)(DEFAULT_INITIAL_CAPACITY *
DEFAULT_LOAD_FACTOR);
    table = new Entry[DEFAULT_INITIAL_CAPACITY];
    init();
}
```

```
static class Entry<K,V> .....{
    final K key;
    V value;
    Entry<K,V> next;
    int hash;
    Entry(int h, K k, V v, Entry<K, V> n) {
        value = v;
        next = n;
        key = k;
        hash = h;
    }
}
```

```

public V put(K key, V value) {
    if (key == null)
        return putForNullKey(value);
    int hash = hash(key.hashCode());
    int i = indexFor(hash, table.length);
    for (Entry<K,V> e = table[i]; e != null; e = e.next) {
        Object k;
        if (e.hash == hash && ((k = e.key) == key ||
key.equals(k))) {
            V oldValue = e.value;
            e.value = value;
            e.recordAccess(this);
            return oldValue;
        }
    }
    modCount++;
    addEntry(hash, key, value, i);
    return null;
}

static int indexFor(int h, int length) {
    return h & (length-1);
}

void addEntry(int hash, K key, V value, int bucketIndex) {
    Entry<K, V> e = table[bucketIndex];
    table[bucketIndex] = new Entry<>(hash, key, value, e);
    if (size++ >= threshold)
        resize(2*table.length);
}

```

Why does Java HashMap use a power-of-two as the length of its array?

Analysis of indexFor and hash methods

Can you see how the HashMap class tries to resolve collisions (open addressing or separate chaining)?

Let's try to use the HashMap for a small application, word frequency counter!

Major methods of the HashMap in java are:

- put(key, value): associates the value with the key in the map
- get(key): returns the value of the key or null.
- keySet(): returns a set view of the keys in the map
- values(): returns a collection view of the values in the map

For keySet() and values() methods, we can use an Iterator to traverse the returned values.

A simple application: implementing the frequency count of words

```

/*
 * You can try to tune initial capacity and load factor.
 * default values are 16 and 0.75 respectively.
 */
Map<String, Integer> freqOfWords = new HashMap<String,
Integer>(10, 0.65f);

String[] words = "coming together is a beginning keeping together
is progress working together is success".split(" ");

for (String word : words) {
    Integer frequency = freqOfWords.get(word);
    if (frequency == null) {
        frequency = 1;
    } else {
        frequency++;
    }
    freqOfWords.put(word, frequency);
}

```

will get length 16 (2^4 , we use the power of 2 in HM)

How do we iterate all the words and print each per one line?

```
Iterator<String> itr = freqOfWords.keySet().iterator();  
while (itr.hasNext()) {  
    System.out.println(itr.next());  
}
```

Or

```
for (String word : freqOfWords.keySet()) {  
    System.out.println(word);  
}
```

HashSet

Java provides HashSet class which is a regular set in which all objects are distinct.

HashSet's major methods are:

- add(key) : adds a key to the set.
- contains(key) : returns true if the set has that key
- iterator() : returns an iterator over the elements

HashSet class implements the Set interface **using HashMap instance**.

Look at some simplified source code snippets of the HashSet class.

```
public class HashSet<E> implements Set<E> {
    private ... HashMap<E, Object> map;
    // Dummy value to associate with an Object in the backing Map
    private static final Object PRESENT = new Object();

    public HashSet() {
        map = new HashMap<>();
    }

    public boolean add(E e) {
        return map.put(e, PRESENT) == null;
    }
}
```

What do you see here?

It creates a HashMap instance and it uses HashMap's put method to add a new element but with dummy object as its value.

And, it adds, or puts, a new element only if it is not already present in the map.

How does it check an object is a duplicate or not?

It stores and retrieves elements by using a hash function that converts elements into an integer.

Let's try to use the HashSet for a small application, distinct word counter!

```
Set<String> distinctWords = new HashSet<String>();
String[] words = "coming together is a beginning keeping together
is progress working together is success".split(" ");

for (String word: words) {
    distinctWords.add(word);
}

System.out.println("There are " + distinctWords.size() + "
distinct words.");
System.out.println("They are: " + distinctWords);
```

How would you iterate and print words one per line?